第九章 电磁感应

electromagnetic induction

§1 法拉第电磁感应定律

§2 动生电动势

§3 感生电动势 感生电场

§4 自感 互感现象

§5 磁场能量

§1 法拉第电磁感应定律

Faraday law of electromagnetic induction

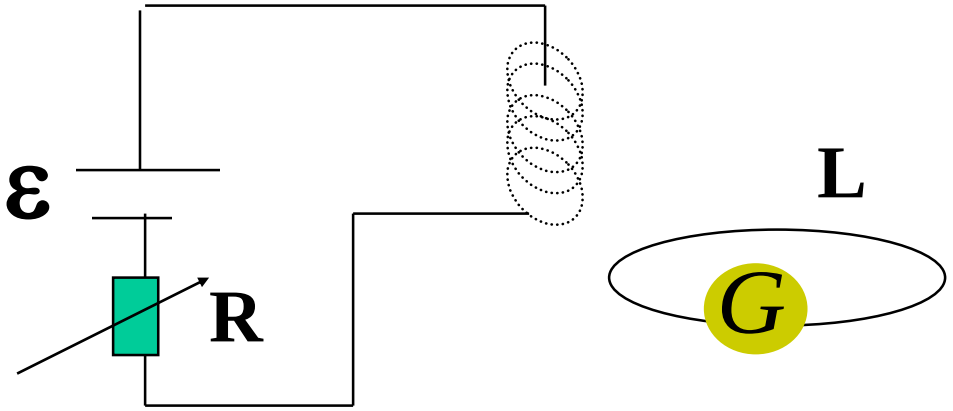
奥斯特　　电流磁效应

对称性 → 磁的电效应?

摸索了十年

变化

一. 现象



1）回路 L 运动
2）场源运动
3）都静止，但 *R* 变化

三种情况均可使电流计指针摆动

二. 规律

1. 法拉第电磁感应定律

induction emf

前面三种情况共同特征是穿过回路的磁通量发生变化

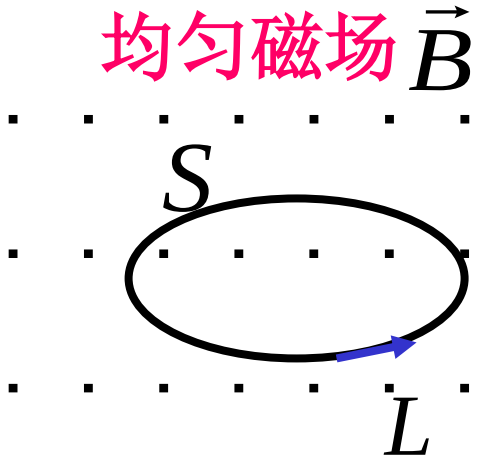
$$\boxed{\varepsilon_i = -\frac{d\phi}{dt}}$$

- 电动势方向与绕行方向一致时为正
- 磁场与绕行方向成右手螺旋时规定磁通量为正

例1 均匀磁场 $\vec{B}$ $\quad \dfrac{dB}{dt} > 0$

求：面积S边界回路中的电动势

解：取一绕行方向



$$\phi = BS$$

$$\varepsilon_i = -\frac{d\phi}{dt} = -\frac{dB}{dt}S < 0$$

取相反绕行方向，结果相同

2. 楞次定律 Lenz law

闭合回路中感应电流的方向，总是使它所激发的磁场来阻止引起感应电流的磁通量的变化。

3. 磁链 magnetic flux linkage

对于N匝串联回路 每匝中穿过的磁通分别为

$$\phi_1, \quad \phi_2, \cdots, \quad \phi_N$$

磁链 $$\boxed{\psi = \sum_i \phi_i}$$

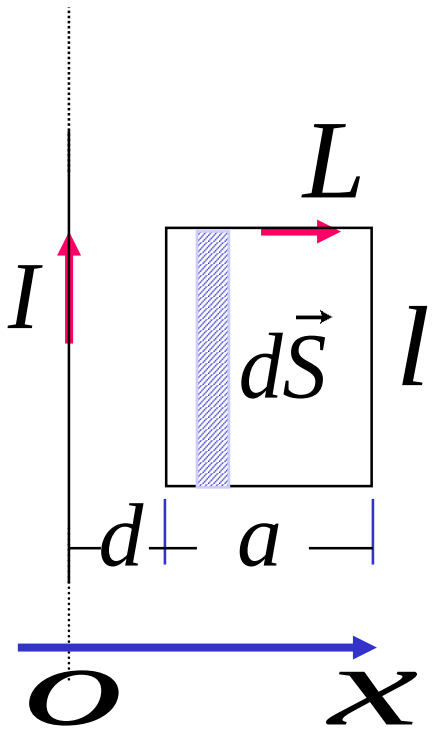
$$\boxed{\varepsilon_i = -\frac{d\psi}{dt}}$$

例2：直导线通交流电 置于磁导率为μ的介质中

求：与其共面的N匝矩形回路中的感应电动势

已知 $I = I_0 \sin \omega t$ 向上为正， 如图所示

$$I_0, \omega > 0$$



解： 回路L方向如图

建坐标系如图

在任意坐标处取一面元 $d\vec{S}$

$$\psi = N\phi = N\int_S \vec{B}\cdot d\vec{S}$$

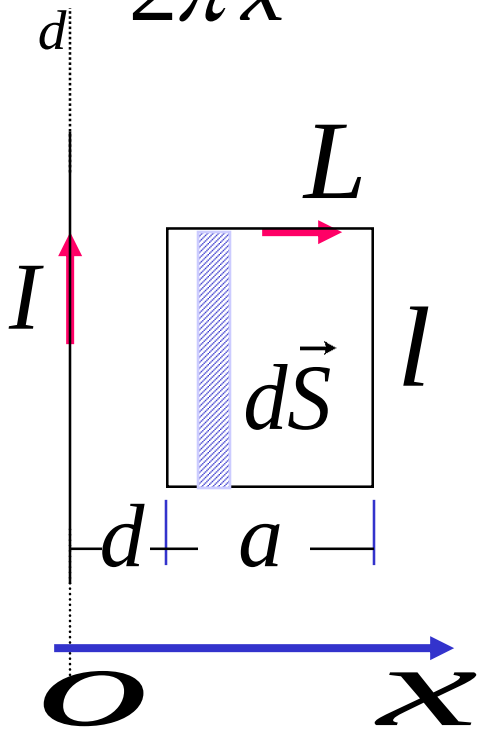
$$\psi = N\phi = N\int_S \vec{B}\cdot d\vec{S} = N\int_S BdS = N\int_d^{d+a} \frac{\mu I}{2\pi x} l dx$$

$$= \frac{N\mu I l}{2\pi}\ln\frac{d+a}{d}$$

$$= \frac{\mu N I_0 l}{2\pi}\sin\omega t\ \ln\frac{d+a}{d}$$

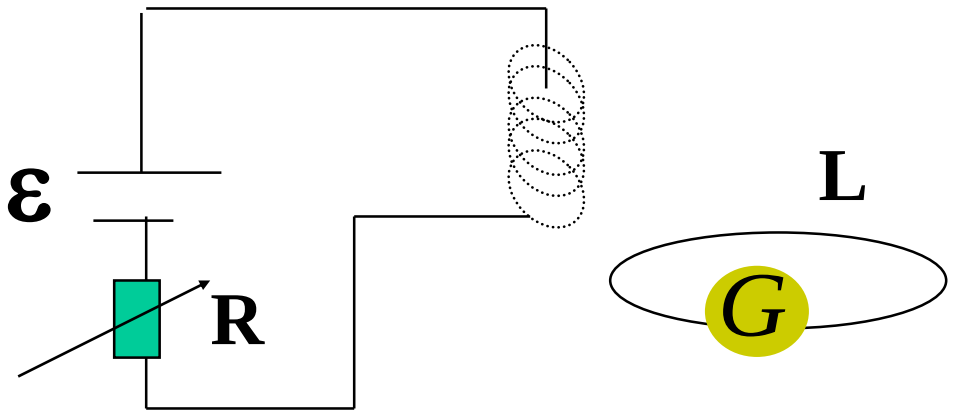
$$\varepsilon_i = -\frac{d\psi}{dt}$$

$$= -\frac{\mu_0\mu_r N I_0 l\omega}{2\pi}\cos\omega t\ln\frac{d+a}{d}$$



交变的
电动势

三. 感应电动势的机理



1）回路 L 运动
2）场源运动
3）都静止，但 *R* 变化

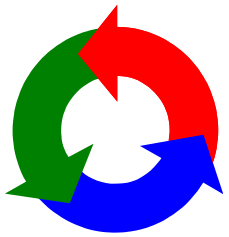
- 动生电动势 motional emf (1)
- 感生电动势 induced emf (2、3)

动生电动势的非静电力?

感生电动势的非静电力?

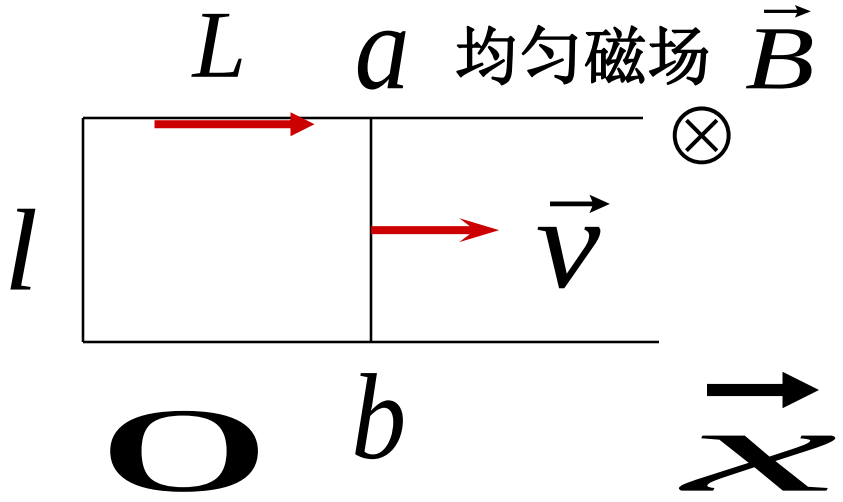
动生电动势:导体(自由电荷)动，运动电荷受力,洛仑兹力?

感生电动势:导体静止，静止电荷受力,必是电场力,有非静电场



§2 动生电动势

典型装置

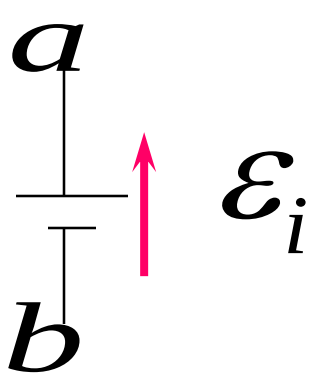


设回路L方向如图

$$\phi = Blx(t)$$

$$\varepsilon_i = -\frac{d\phi}{dt} = -Bl\frac{dx}{dt}$$

$$= -Blv$$



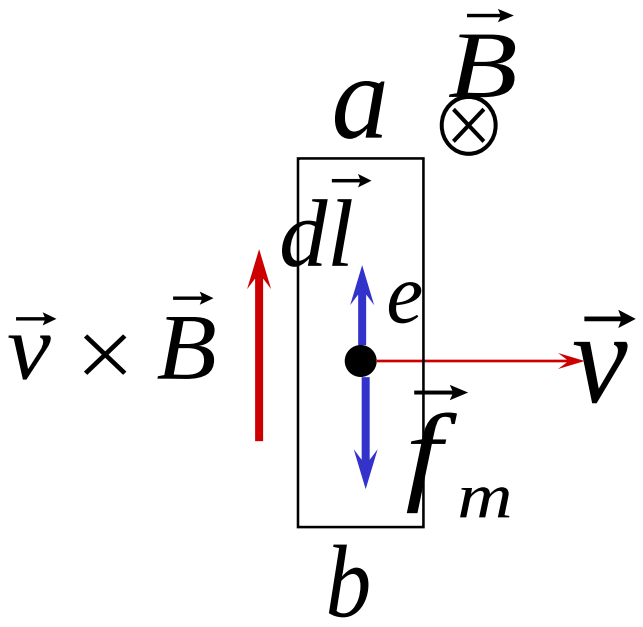
负号说明电动势方向与所设方向相反

运动电荷受驱动产生电动势

非静电力——洛仑兹力

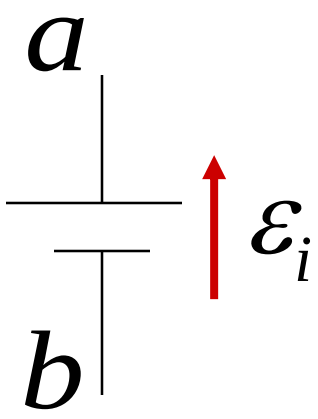
$$\boxed{\vec{f}_m = q\vec{v} \times \vec{B}}$$

$$\boxed{\vec{E}_K = \frac{q\vec{v} \times \vec{B}}{q} = \vec{v} \times \vec{B}}$$



$$\varepsilon_i = \int_{(b)}^{(a)} \left(\vec{v} \times \vec{B}\right) \cdot d\vec{l}$$

$$\varepsilon_i = \int_{(ba)} vBdl = vBl \quad > 0$$

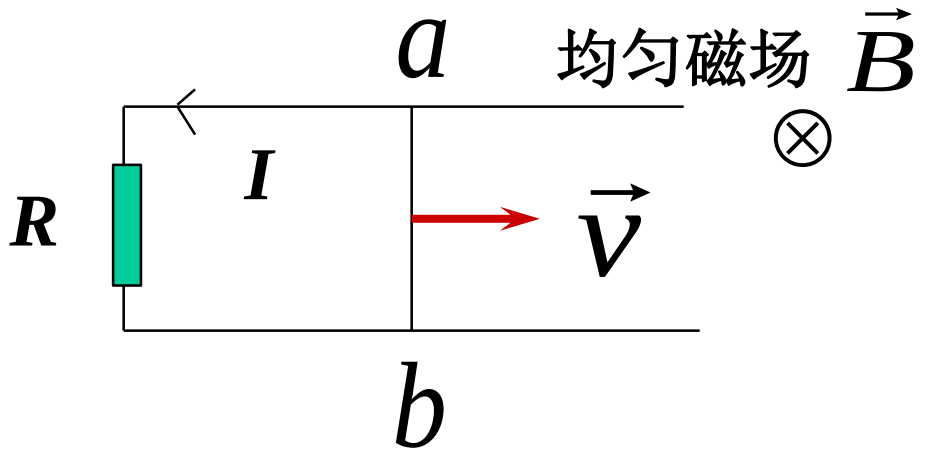


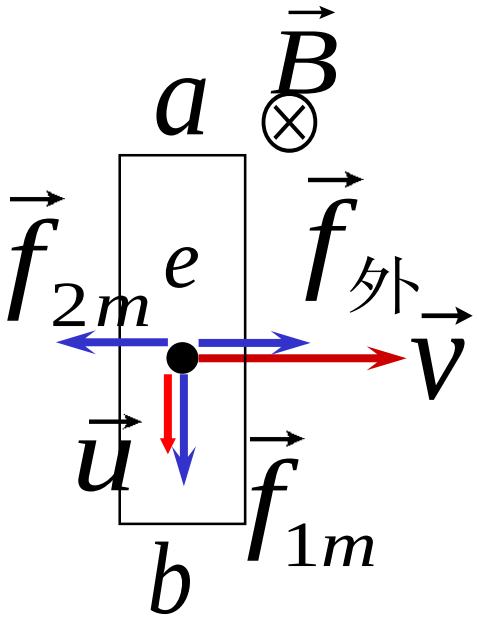
- 磁场中运动的导体

$$d\varepsilon_i = \left(\vec{v} \times \vec{B}\right) \cdot d\vec{l} \qquad \varepsilon_i = \int d\varepsilon_i$$

- 电动势要对电路中载流子做功

动生电动势本质是洛仑兹力，但洛仑兹力不做功，矛盾？





$$\vec{F} = \vec{f}_{1m} + \vec{f}_{2m} = -e(\vec{v} + \vec{u}) \times \vec{B}$$

$$\perp \vec{v} + \vec{u}$$

$$\vec{F} \cdot (\vec{v} + \vec{u}) = 0$$

$$= \vec{f}_{1m} \cdot \vec{u} + \vec{f}_{2m} \cdot \vec{v}$$

$$0 \neq \quad \vec{f}_{1m} \cdot \vec{u} = -\vec{f}_{2m} \cdot \vec{v} \quad = \vec{f}_{外} \cdot \vec{v} \qquad nSl \times$$

$$nSl\ qvB\ u = nSl\ \vec{f}_{外} \cdot \vec{v} \qquad nqu\ S\ vBl = I\varepsilon = \vec{F}_{外} \cdot \vec{v}$$

外力作功转换为对电路作功

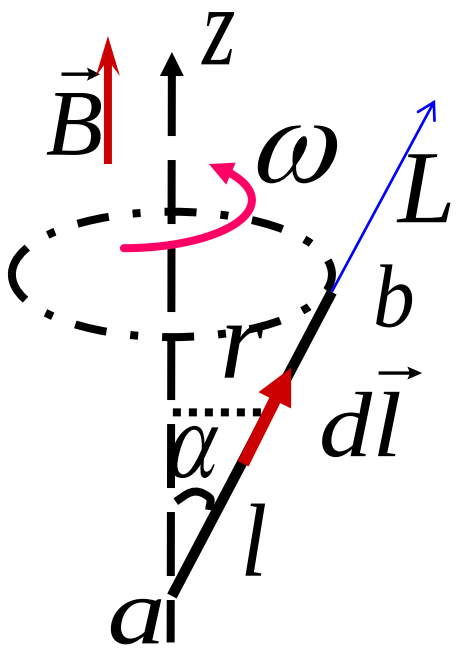
例3 在空间均匀的磁场中 $\vec{B} = B\hat{z}$

导线ab绕 Z 轴以ω匀速旋转

导线ab与Z轴夹角为α

设 $\overline{ab} = L$

求：导线ab中的电动势



解：建坐标如图 在坐标l处取 dl

该段导线运动速度垂直纸面向内

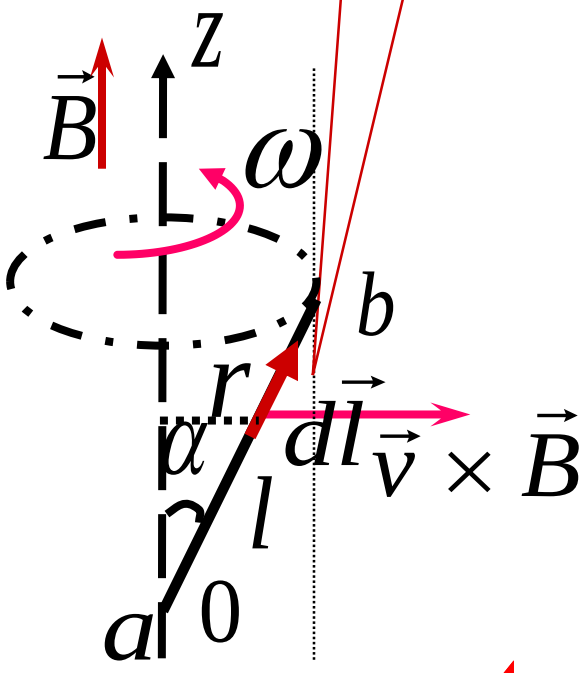
运动半径为r

$$\left|\vec{v}\times\vec{B}\right| = vB = \omega rB = \omega lB\sin\alpha$$

$$\theta = \frac{\pi}{2} - \alpha$$

$$d\varepsilon_i = (\vec{v}\times\vec{B})\cdot d\vec{l} = vBdl\cos\theta$$

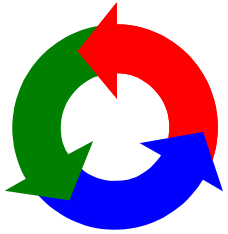
$$= B\omega\sin^2\alpha\ ldl$$



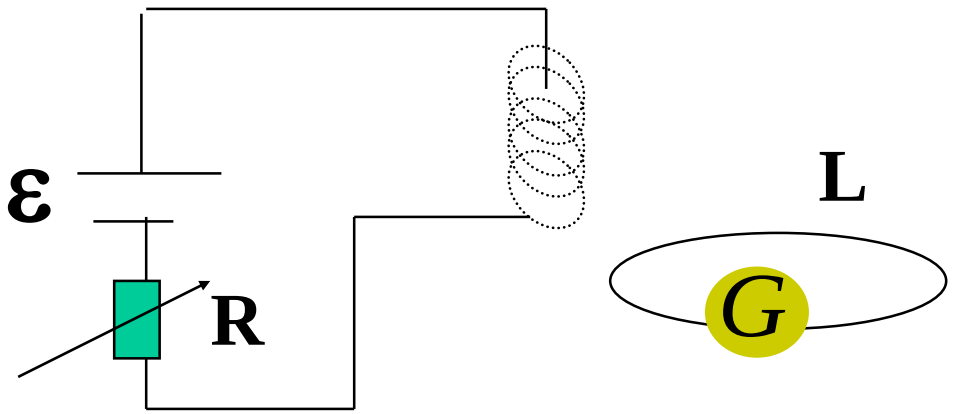
$$\varepsilon_i = \int d\varepsilon_i = B\omega\sin^2\alpha\int_0^L ldl$$

$$= \frac{B\omega L^2}{2}\sin^2\alpha > 0$$

方向从 $\boldsymbol{a\to b}$



§3 感生电动势 感生电场



2）场源运动

3）都静止，但 R 变化

回路不动, 磁场随时间变化

$$\vec{B} = \vec{B}(\vec{r}, t)$$

$$\phi = \iint_S \vec{B} \cdot d\vec{S}$$

$$\boxed{\varepsilon_i = -\frac{d\psi}{dt}} \Rightarrow \boxed{\varepsilon_i = -\iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}}$$

电动势 $$\varepsilon = \oint_L \vec{E}_{非} \cdot d\vec{l}$$

$$\oint_L \vec{E}_{非} \cdot d\vec{l} = -\iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

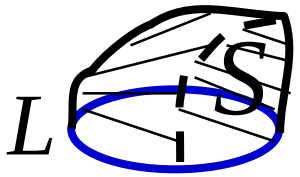
静止电荷受驱动, 一定是电场 （非静电场）

- 感生电场是法拉第电磁感应定律的另一机理

  只要有磁场的变化， 就存在感生电场。

一. 感生电场的性质

$$\boxed{\oint_L \vec{E}_{感生} \cdot d\vec{l} = -\iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}}$$



S是以L为边界的任意曲面

环量不为零　　非保守场

非电荷产生

$$\boxed{\oiint_S \vec{E}_{感生} \cdot d\vec{S} = 0}$$

无源场　涡旋场

二. 感生电场的计算

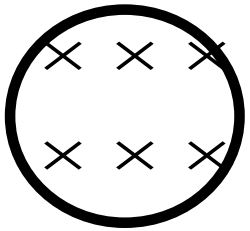
$$\boxed{\oint_L \vec{E}_{感生} \cdot d\vec{l} = -\iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}}$$

$\vec{E}_{感生}$具有某种对称性才有可能计算出来

例4 空间均匀的磁场限制在半径为 R 的圆柱内， $\vec{B}$ 的方向平行柱轴

且有 $\boxed{\dfrac{dB}{dt} = c}$

$\vec{B}(t)$



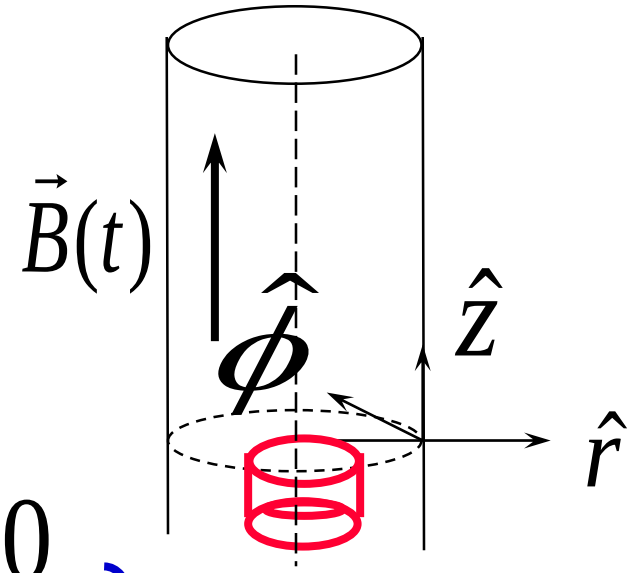
求：$E_{感生}$ 分布

感生电场对称性的分析

限制在圆柱内的空间
均匀的变化磁场

建柱坐标系

$$\vec{E}_{感生} = E_r\hat{r} + E_\phi\hat{\phi} + E_z\hat{z}$$



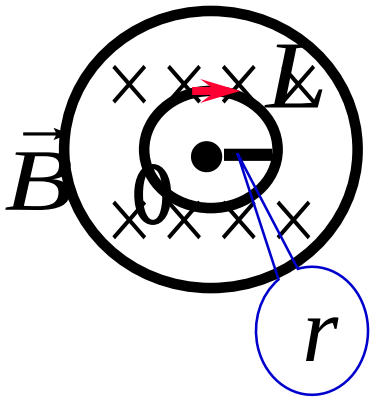
对称性(平移/反射) → $E_z = 0$

轴对称, 作正柱面

$$\oiint_S \vec{E}_{感生} \cdot d\vec{S} = 0$$ → $E_r = 0$

$$\boxed{\vec{E}_{感生} = E_\phi\hat{\phi}}$$

解：设场点距轴心为r，根据对称性，取以o为心，过场点的圆周环路 L



$$\oint_L \vec{E}_{感生} \cdot d\vec{l} = E_{感生} 2\pi r$$

由法拉第电磁感应定律

$$= -S\frac{dB}{dt}$$

$$\boxed{E_{感生} = -\frac{S}{2\pi r}\frac{dB}{dt}}$$

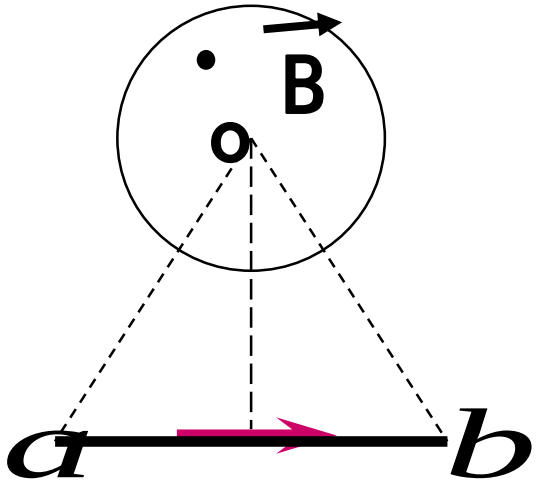
$r < R \quad S = \pi r^2 \longrightarrow$

$$\boxed{E_{感生} = -\frac{r}{2}\frac{dB}{dt}}$$

$r > R \quad S = \pi R^2 \longrightarrow$

$$\boxed{E_{感生} = -\frac{R^2}{2r}\frac{dB}{dt}}$$

例5 磁力线均匀限制在圆柱体空间内 $\dfrac{dB}{dt}=c$



求：ε_{ab}

解：补上半径 ***oa bo***

$$\varepsilon_i=\varepsilon_{oa}+\varepsilon_{ab}+\varepsilon_{bo}=-\frac{d\varphi}{dt}$$

$\vec{E}_{感生}\perp R$ $\quad \varepsilon_{感生}=\int_{(R)}\vec{E}\cdot d\vec{l}=0$ $\quad \varepsilon_{oa}=0$ $\quad \varepsilon_{bo}=0$

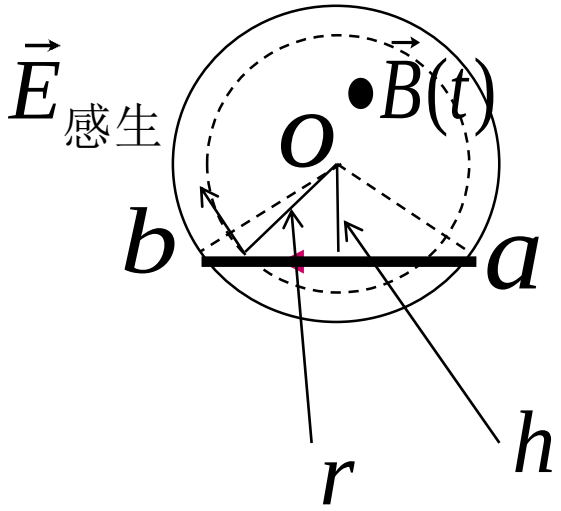
$$\varepsilon_{ab}=-\frac{d\phi}{dt}$$

$$\boxed{\varepsilon_{ab}=-S_{扇形}\frac{dB}{dt}}$$

例6 求线段*ab*线上的感生电动势

解：补上两个半径*oa*和*bo*

$$\boxed{\varepsilon_{ba} = -S_{\Delta}\frac{dB}{dt}}$$

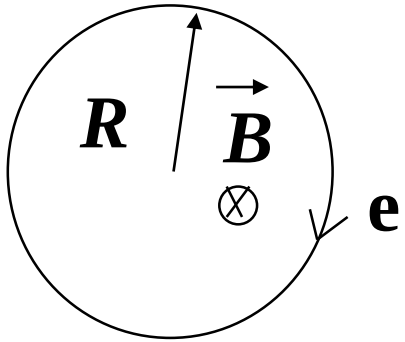


$$E_{感生} = -\frac{r}{2}\frac{dB}{dt}$$

$$\varepsilon_{ab} = \int_a^b \vec{E}_{非}\cdot d\vec{l} = \int_a^b \frac{r}{2}\frac{dB}{dt}\frac{h}{r}dl = \frac{h}{2}\overline{ab}\frac{dB}{dt}$$

三. 感生电场的应用

1. 电子感应加速器



1947年世界第一台 70MeV

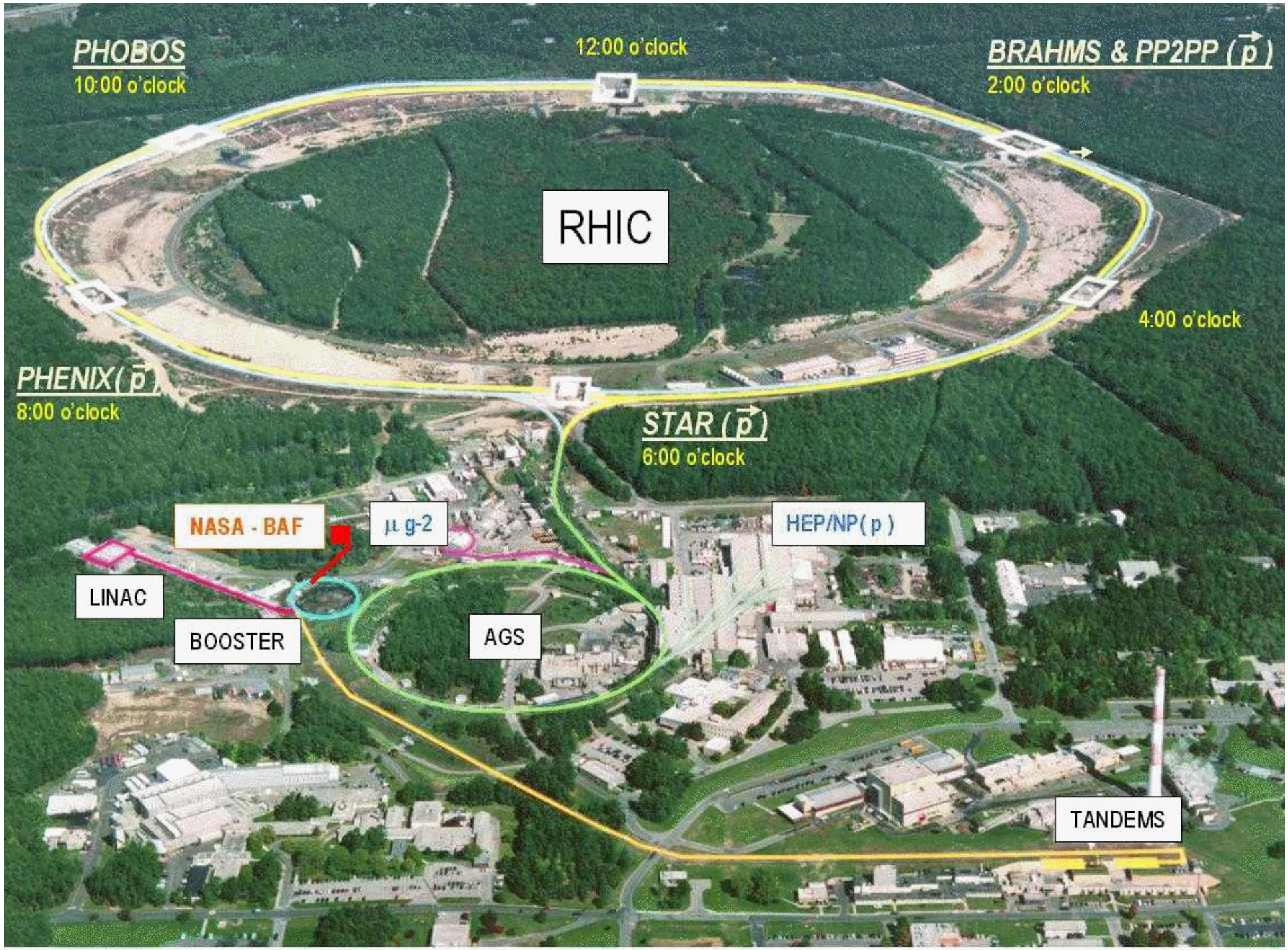
$$\gamma \approx 140$$

$$v \approx 0.9999745c$$

$$eB_R\upsilon = m\frac{\upsilon^2}{R} \rightarrow \upsilon = \frac{e}{m}B_R R$$

$$\frac{d}{dt}(m\upsilon) = -eE_{感生} = \frac{e}{2\pi R}\frac{d\Phi}{dt} \xrightarrow[\upsilon=0,\ \Phi=0]{} m\upsilon = \frac{e\Phi}{2\pi R}$$

$$B_R = \frac{\Phi}{2\pi R^2} = \frac{1}{2}\bar{B} \qquad \bar{B} = \frac{1}{\pi R^2}\iint_{r\le R}\vec{B}\cdot d\vec{S}$$

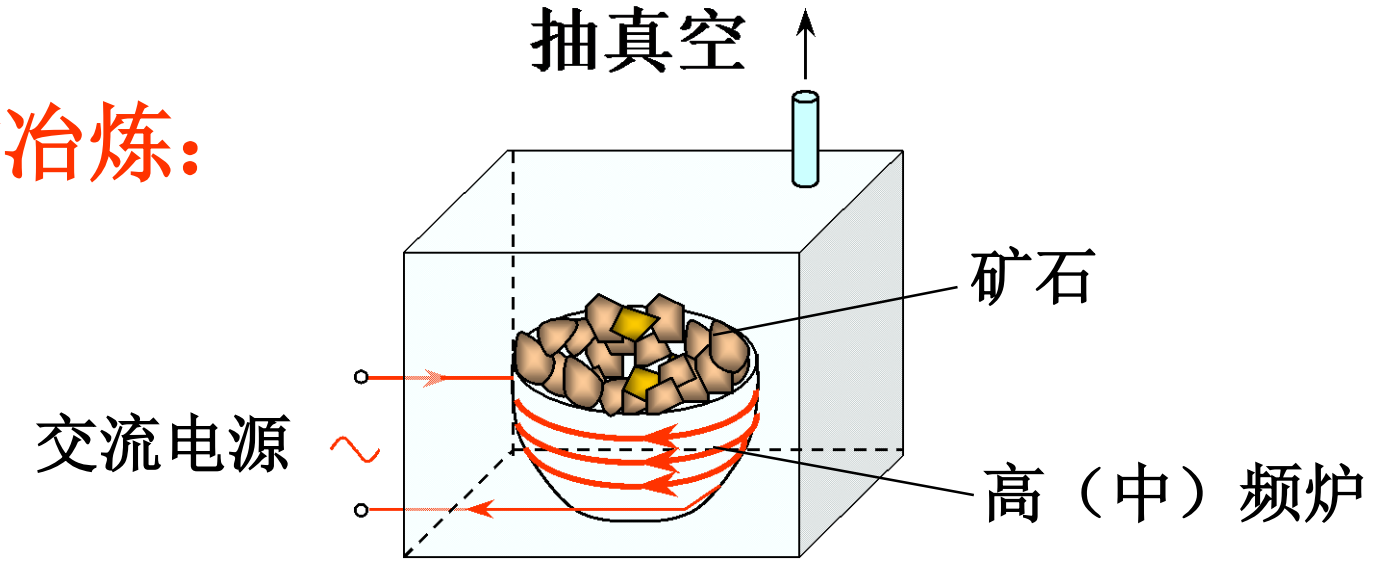

PHOBOS
10:00 o'clock
12:00 o'clock
BRAHMS & PP2PP (p)
2:00 o'clock
RHIC
4:00 o'clock
PHENIX (p)
8:00 o'clock
STAR (p)
6:00 o'clock
NASA - BAF
μ g-2
HEP/NP (p)
LINAC
BOOSTER
AGS
TANDEMS

2. 涡流
（eddy current）

大块导体中的感应电流称“涡流”。

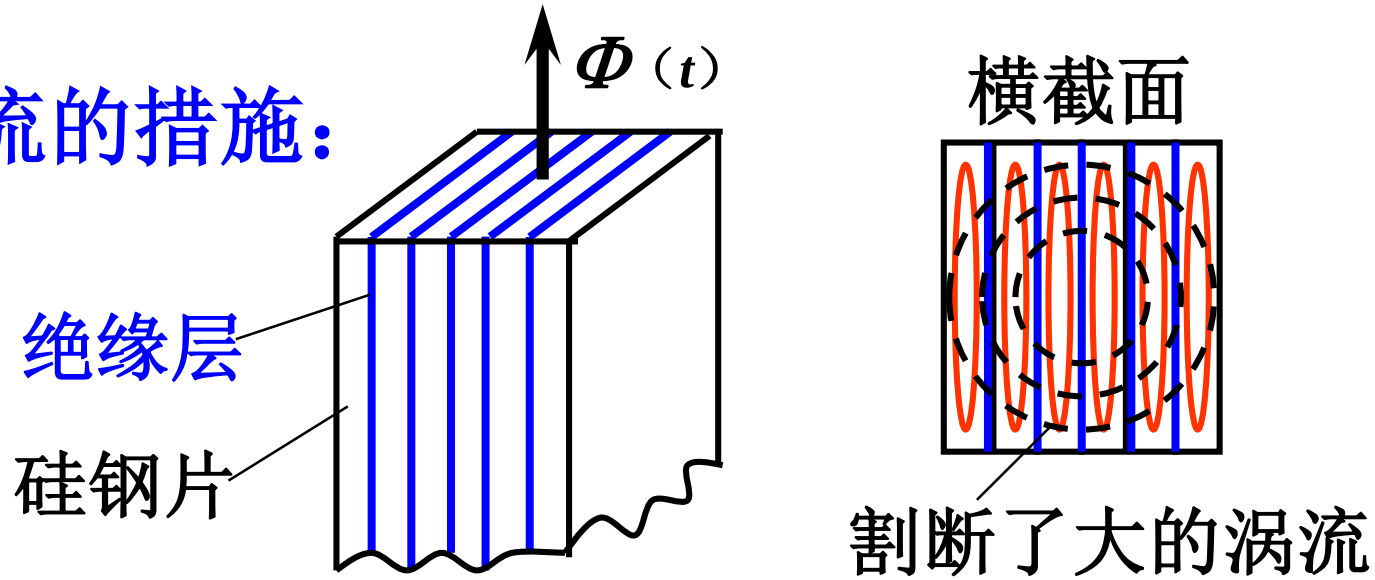
▲热效应

电磁冶炼：

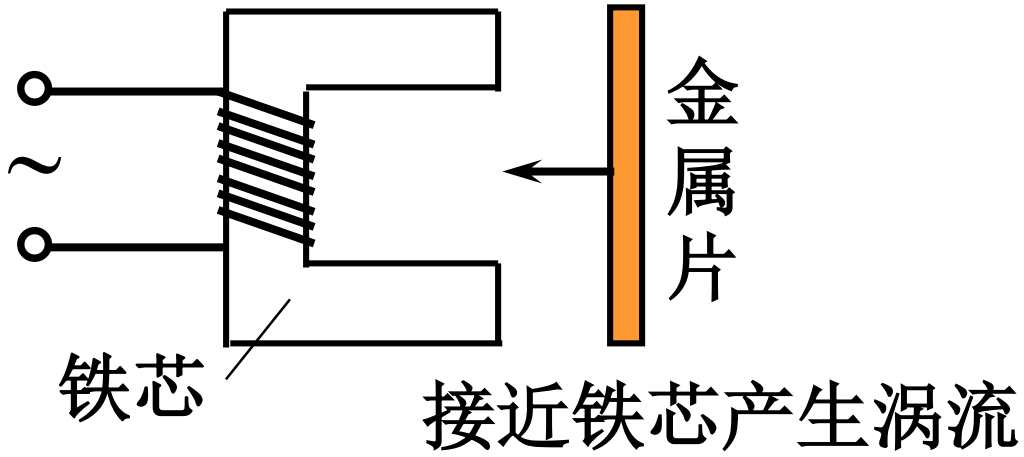


电磁淬火

减小涡流的措施：

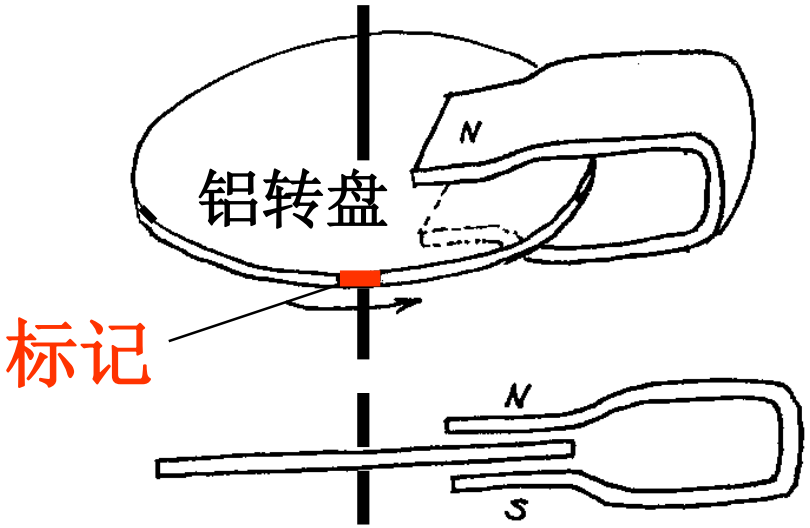


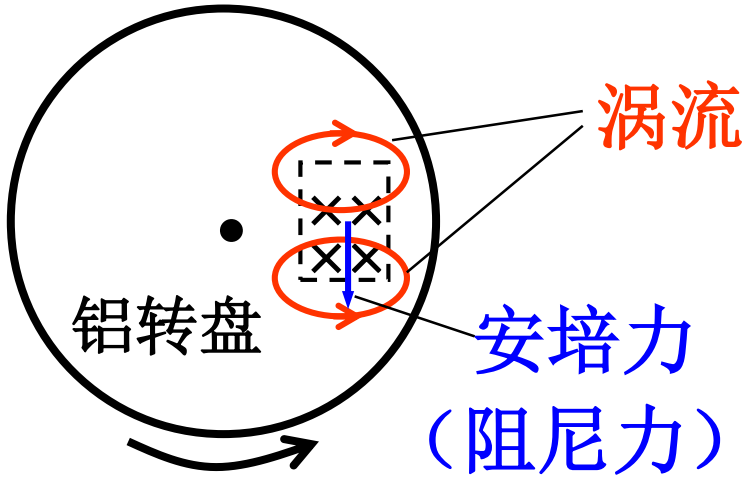
▲电磁效应（用于控制）



▲机械效应

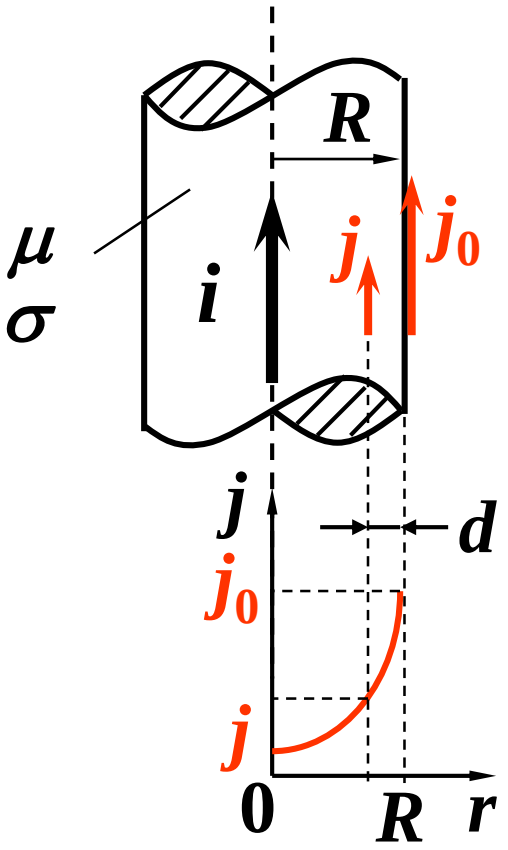
电磁阻尼：



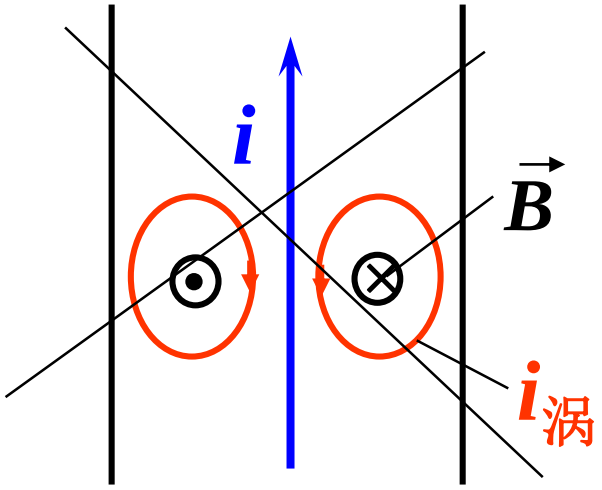


*（高频）趋肤效应 （skin effect）

高频电流通过导线时，越靠表面电流密度越大。



$$d = d_S \text{ 时，} j = \frac{j_0}{e} \approx 0.37\, j_0$$

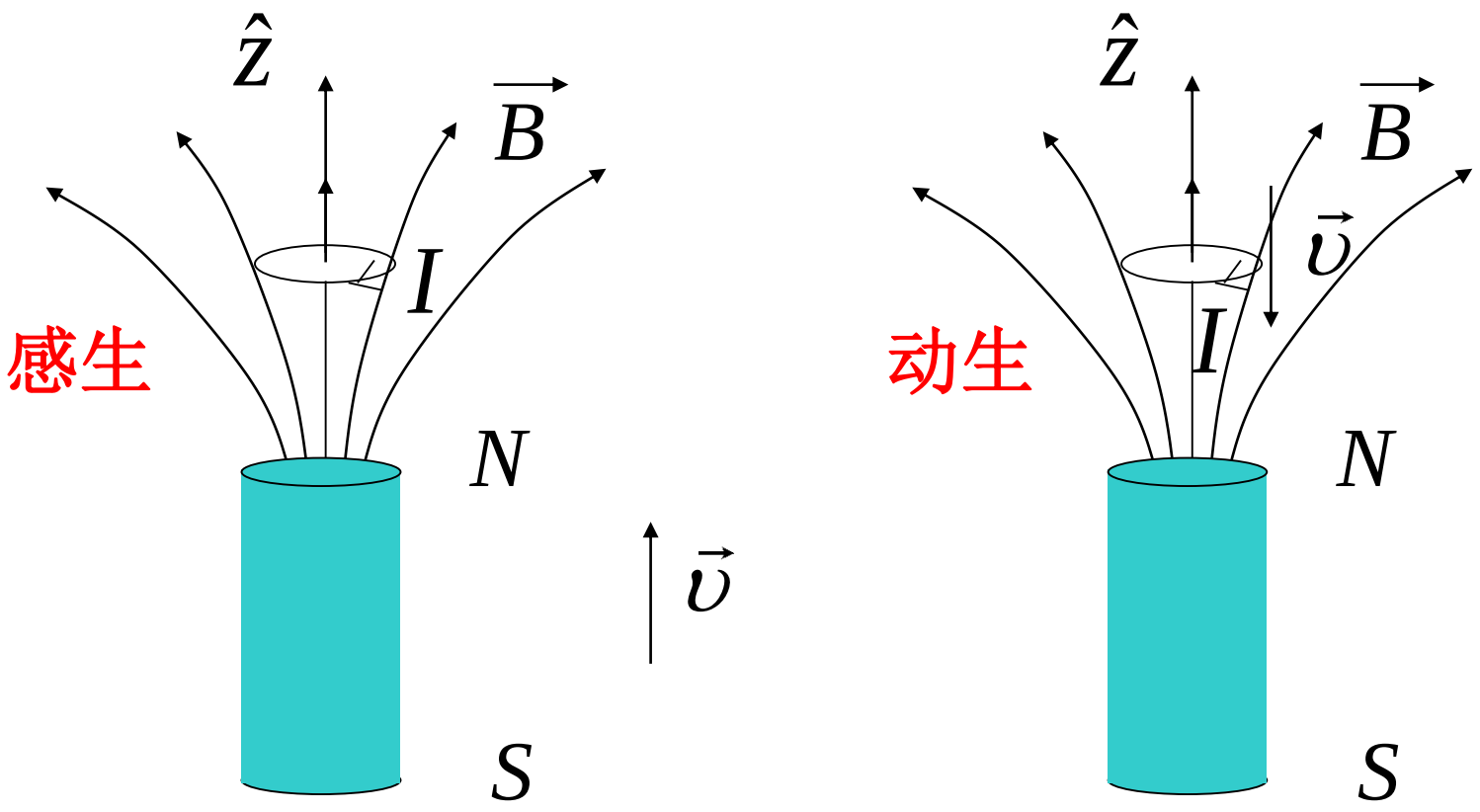


i、$\vec{B}$、$i_{涡}$ 的正方向

简单用普通物理直观解释行不通!

四. 动生电动势 和 感生电动势的相对性

线圈不动 源动　　在源参考系　线圈动

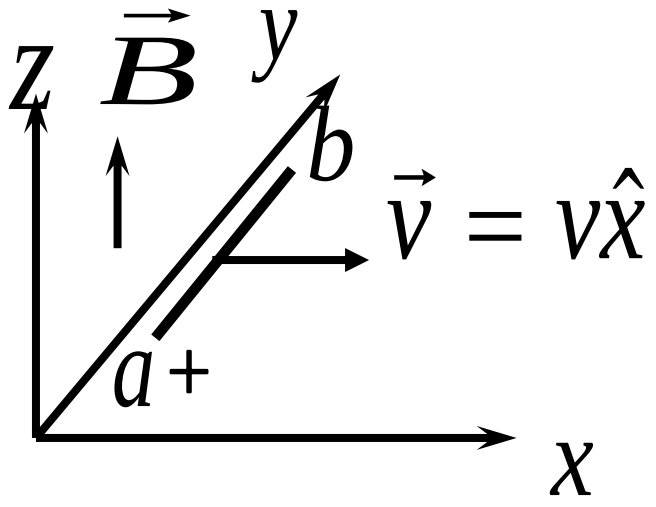


静止电荷受电场力 <=> (换参考系)运动电荷在磁场受力

例7 解释实验现象

地面上S系导体细棒 ab

$$\vec{B} = B_z \hat{z} \qquad \vec{v} = v\hat{x}$$



电子运动受磁场力

$$q\vec{v} \times \vec{B} = qvB(-\hat{y}) = f_y \hat{y}$$

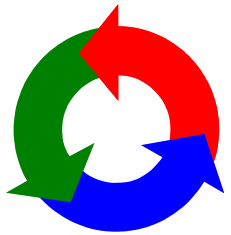
站在导体棒上S`系　　电子不运动不受磁力

$$\gamma f_y \hat{y} = \gamma q v B_z (-\hat{y})$$

受电场力

$$= qE'_y \hat{y}$$

$$\boxed{E'_y = -\gamma v B_z}$$



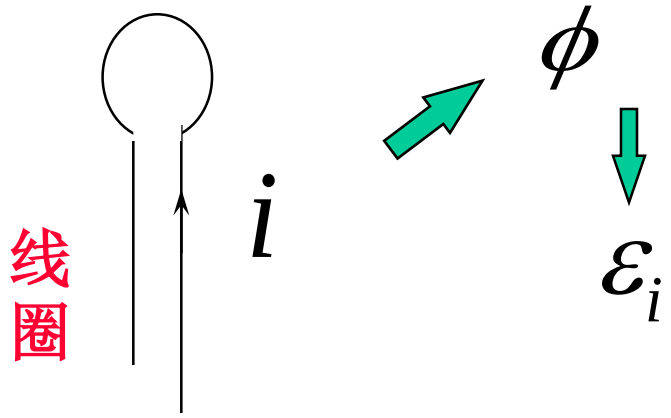
§4 自感　互感现象

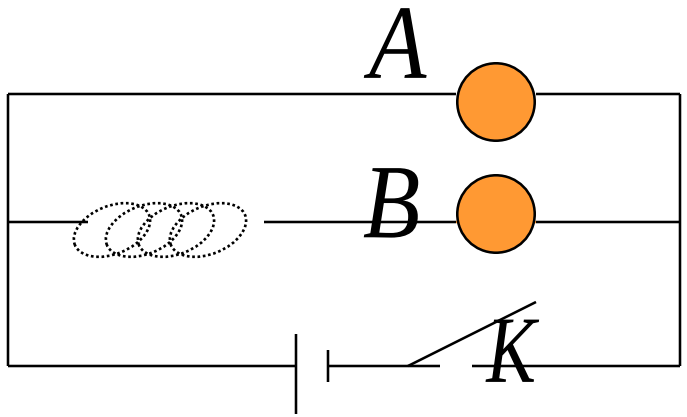
实际线路中的感生电动势问题

一. 自感现象　自感系数　self-inductance

反抗电流变化的能力

(电惯性)





***K*合上 灯泡*A*先亮 *B*晚亮**

线路中的电流的变化，在自身线路中产生感应电流的现象——自感现象

自感系数的定义

毕-萨-拉+叠加

非铁磁质 $I \rightarrow \psi \propto I \rightarrow \psi = LI$

由法拉第电磁感应定律

$$\varepsilon_i = -\frac{d\psi}{dt} = -L\frac{dI}{dt}$$

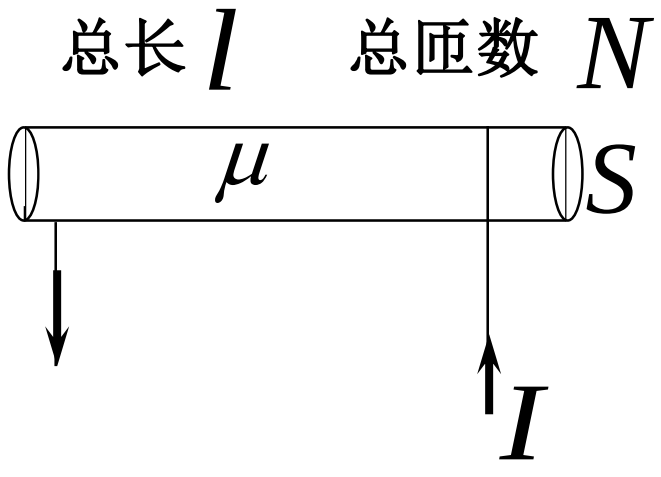
$$\boxed{L = \frac{\psi}{I}}$$

$$L = -\frac{\varepsilon_i}{dI/dt}$$

单位电流的变化对应的感应电动势 普遍定义

例8：求长直螺线管的自感系数

几何条件如图



解：设通电流 I

$$\boxed{B = \mu \frac{N}{l} I}$$

$$\psi = N\phi = NBS$$

自感器件

$$\boxed{L = \frac{\psi}{I} = \frac{\mu N^2 S}{l}}$$

几何条件

介质

*二、暂态过程

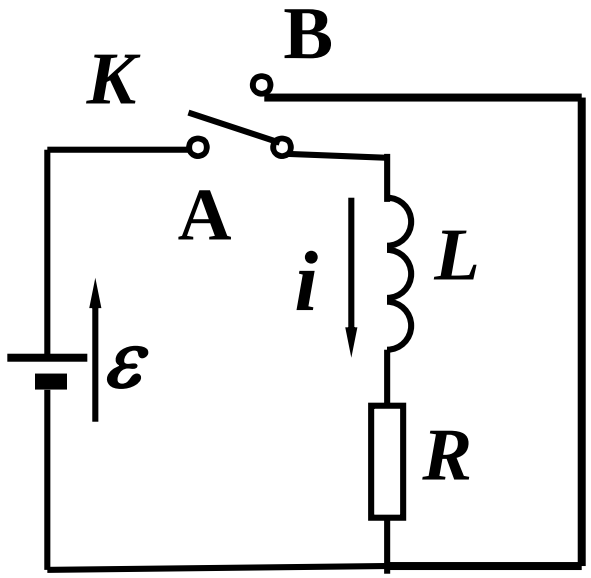
电惯性

R-L

自感线圈中 $\varepsilon_L \neq \infty \rightarrow \dfrac{\mathrm{d}i}{\mathrm{d}t} \neq \infty \rightarrow i$不能突变

由楞次定律得知，i 的变化受到 ε_L 的阻碍，

∴L对变化电流有感抗，但对直流电流畅通。

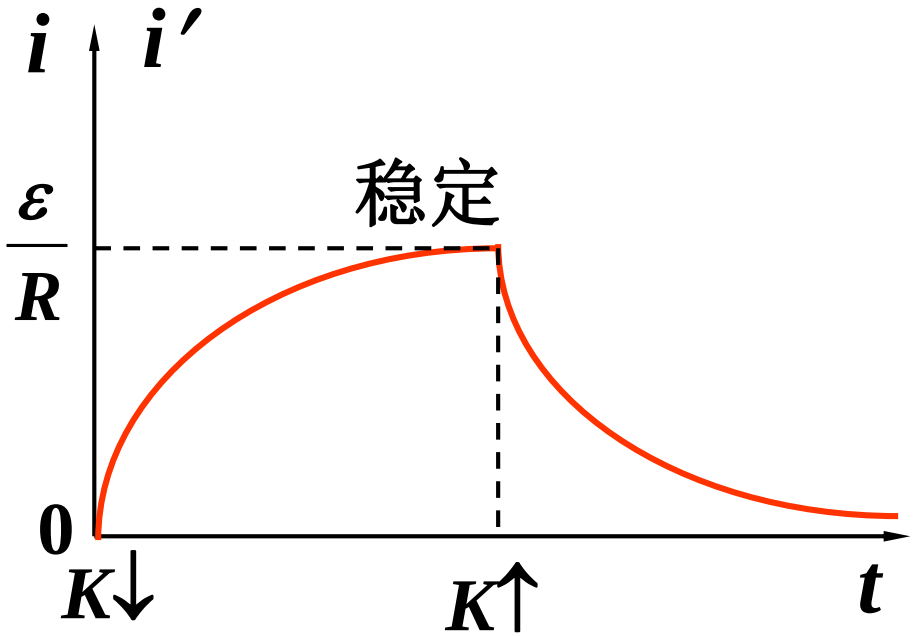


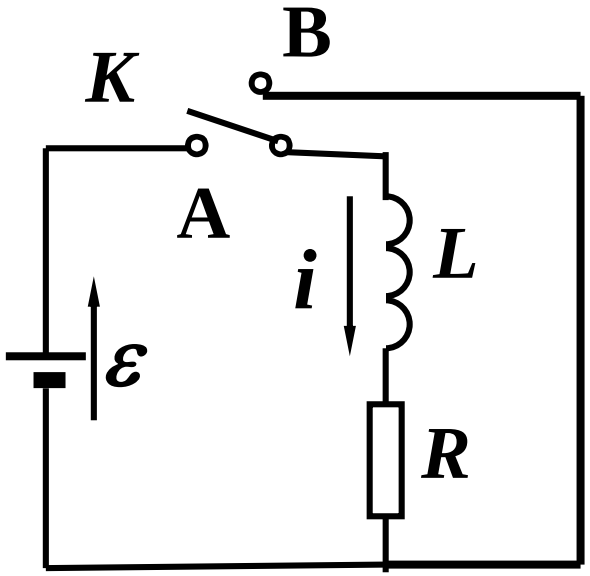
(K-A) $$\varepsilon = iR + L\frac{di}{dt}$$

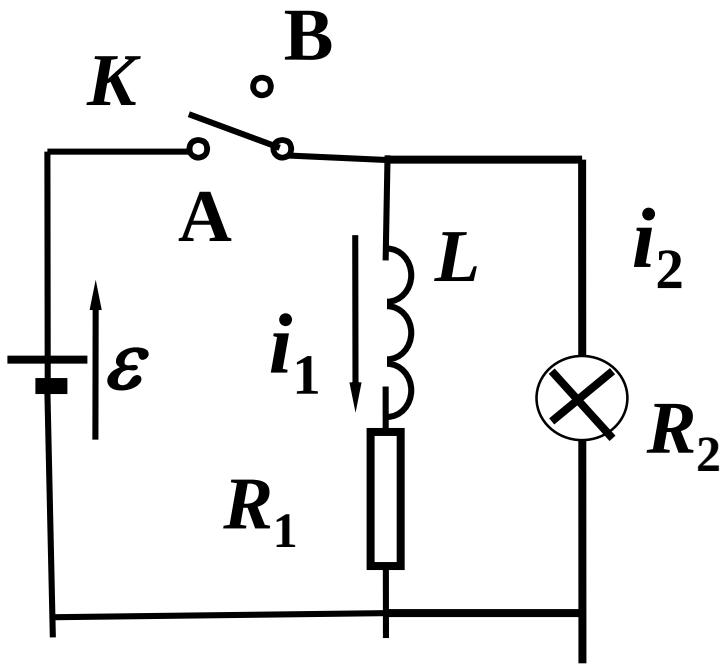
(K-B) $$0 = iR + L\frac{di}{dt}$$

$K\downarrow$： $i=\dfrac{\varepsilon}{R}(1-e^{-t/\tau})$,　　$K\uparrow$： $i'=\dfrac{\varepsilon}{R}\,e^{-t/\tau}$

$$\tau=\frac{L}{R}$$ — 时间常数







(K-A)

$$R_2 > R_1$$

$$i_1 > i_2$$

(K-B)

i_1 不能马上变零,
流过 R_2

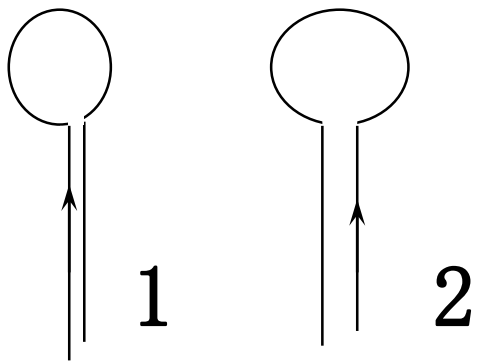
三. 互感现象　互感系数　mutual induction

第一个线圈内电流的变化，引起线圈2内的电动势

非铁磁质

$$\boxed{\psi_{21} = M_{21} I_1}$$

毕-萨-拉+叠加



同样有

$$\boxed{\psi_{12} = M_{12} I_2}$$

互感系数带符号

由法拉第电磁感应定律有

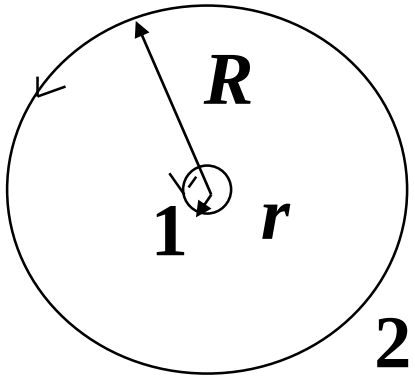
$$\boxed{\varepsilon_{21} = -\frac{d\psi_{21}}{dt} = -M_{21}\frac{dI_1}{dt}}$$

$$\boxed{\varepsilon_{12} = -\frac{d\psi_{12}}{dt} = -M_{12}\frac{dI_2}{dt}}$$

可以证明 $\boxed{M_{21} = M_{12} = M}$ 互感系数

$$\boxed{M = -\frac{\varepsilon_{21}}{dI_1/dt} = -\frac{\varepsilon_{12}}{dI_2/dt} = \frac{\psi_{21}}{I_1} = \frac{\psi_{12}}{I_2}}$$

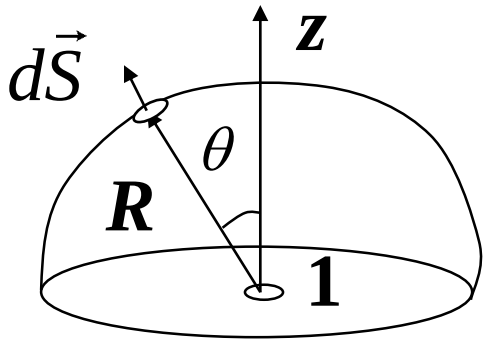
例9 求互感系数



M_{21}

$R >> r$

1) $$M_{21} = \frac{\psi_{21}}{I_1} \qquad \vec{B} = \frac{\mu_0}{4\pi R^3} I_1 \left[-\vec{S}_1 + 3\left(\vec{S}_1 \cdot \hat{r}\right)\hat{r} \right]$$



2

$$\psi_{21} = \iint_S \vec{B} \cdot d\vec{S}$$

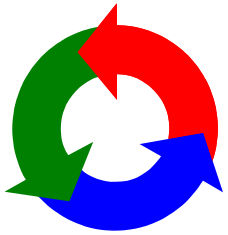
$$d\vec{S} = R^2 d\Omega \hat{r} = R^2 \sin\theta d\theta d\varphi \hat{r}$$

$$\psi_{21} = \iint_S \vec{B} \cdot d\vec{S} = \int_0^{2\pi} \int_0^{\pi/2} \frac{\mu_0}{2\pi R} I_1 S_1 \sin\theta \cos\theta d\theta d\varphi$$

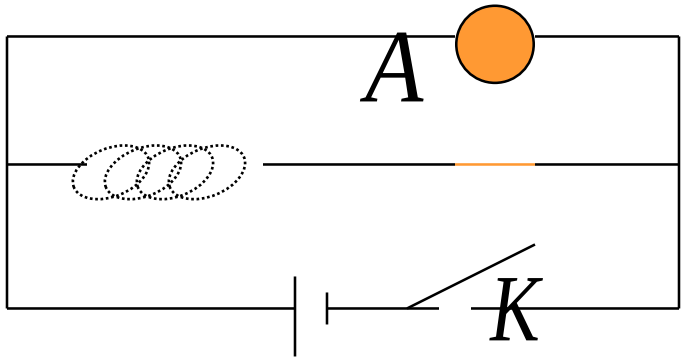
$$\psi_{21} = \frac{\mu_0 I_1 S_1}{2R} \qquad M_{21} = \frac{\psi_{21}}{I_1} = \frac{\mu_0 \pi r^2}{2R}$$

2) 利用

$$\boxed{M_{12} = M_{21}} \qquad \psi_{12} = \frac{\mu_0 I_2 S_1}{2R} \qquad M_{12} = \frac{\psi_{12}}{I_2} = \frac{\mu_0 \pi r^2}{2R}$$

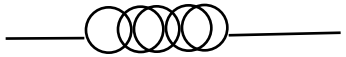


§5　磁场能量



K断开 A 会突闪

电能从哪来？　　储存在电感中的磁能转化



L

电能以磁场能的形式储存下来

克服自感电动势作功

$$W_L = -\int_0^{\infty} I\varepsilon_L dt = \int_0^{\infty} LI\frac{dI}{dt}dt = L\int_0^{I} IdI = \frac{1}{2}LI^2$$

1. 自感磁能

$$\boxed{W_L = \frac{1}{2} L I^2}$$

2. 互感磁能

克服互感电动势作功

$$W_M = -\int_0^{\infty} I_1 \varepsilon_{12} dt - \int_0^{\infty} I_2 \varepsilon_{21} dt = \int_0^{\infty} I_1 M_{12} \frac{dI_2}{dt} dt + \int_0^{\infty} I_2 M_{21} \frac{dI_1}{dt} dt$$

$$= \int_0^{I_1, I_2} M(I_1 dI_2 + I_2 dI_1) = M I_1 I_2$$

$$\boxed{W_M = M I_1 I_2}$$

3. N个线圈系统

两个线圈 $$W_m = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 + M_{12} I_1 I_2$$

N个线圈 $$\boxed{W_m = \frac{1}{2}\sum_i L_i I_i^2 + \frac{1}{2}\sum_{i \neq j} M_{ij} I_i I_j}$$

M_{ij} 互感系数是带符号的

$$\boxed{W_m = \frac{1}{2}\sum_i \left(L_i I_i + \sum_{j \neq i} M_{ij} I_j \right) I_i = \frac{1}{2}\sum_i \psi_i I_i}$$

类比静电能 $$W_e = \frac{1}{2}\sum_i q_i U_i$$

4. 磁矩在磁场中的能量

$$W_m = M_{12} I_1 I_2 = \vec{B}_{12} \cdot \vec{S}_1 I_1 = \vec{p}_m \cdot \vec{B}_{12}$$

考虑所有线圈产生的磁场

$$\boxed{W_m = \vec{p}_m \cdot \vec{B}}$$

这是考虑了磁矩大小可变化的情况

p_m不变, 在外磁场中能量

$$\boxed{W_m = -\vec{p}_m \cdot \vec{B}}$$

例10 电流分别为 I_1 , I_2 ,

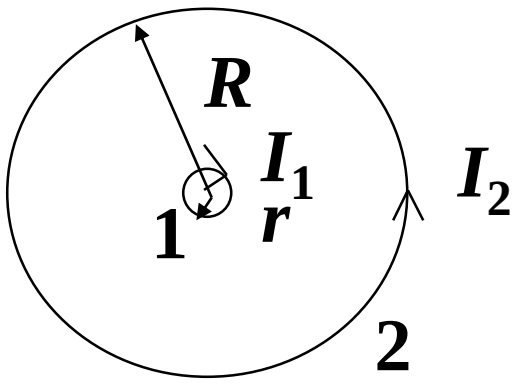
1) 求相互作用能?

2) 固定电流大小, 将小线圈从无限远移来, 外力所做功?

3) 此过程为了维持电流不变, 电源需要作功多少?

4) 能量转换关系如何?

解: 1)



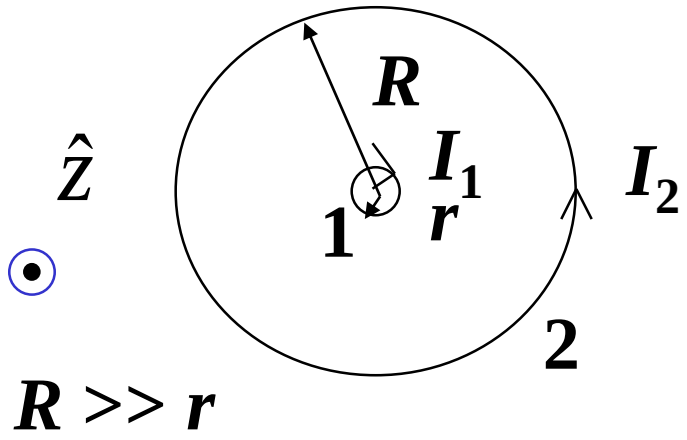
$R >> r$

$$\psi_{12} = MI_2 = -\frac{\mu_0 I_2 S_1}{2R}$$

$$W_m = MI_1 I_2 = -\frac{\mu_0 S_1}{2R} I_1 I_2$$

$$= \vec{m} \cdot \vec{B}$$

2)



$$= -\vec{m}\cdot\vec{B}$$

$$W_{外} = -\int_{\infty}^{0} \vec{F}_m \cdot \hat{z}dz = \int_{\infty}^{0} m\frac{\partial B_z}{\partial z}dz = mB_z(0) = \frac{\mu_0 S_1}{2R} I_1 I_2$$

3. 过程维持电流不变, 电源抵抗互感电动势作功

$$W_1 = -\int_{\infty}^{0} I_1 \varepsilon_{12} dt = \int_{\infty}^{0} I_1 \frac{d(M_{12} I_2)}{dt} dt = I_1 I_2 \int_{\infty}^{0} dM$$

$$W_1 = I_1 I_2 \int_{\infty}^{0} dM = I_1 I_2 (M|_0 - M|_\infty) = M I_1 I_2 = -\frac{\mu_0 S_1}{2R} I_1 I_2$$

$$W_2 = -\int_{\infty}^{0} I_2 \varepsilon_{21} dt = \int_{\infty}^{0} I_2 \frac{d(M_{21} I_1)}{dt} dt = -\frac{\mu_0 S_1}{2R} I_1 I_2$$

$$W_{电源} = W_1 + W_2 = -\frac{\mu_0 S_1}{R} I_1 I_2 = 2\vec{m} \cdot \vec{B}$$

4) 外界作功总和等于系统能量增量

$$W_m - W_\infty = W_{外} + W_{电源}$$

无限远处相互作用能为零

$$W_m = W_{外} + W_{电源} = \vec{m} \cdot \vec{B}$$

5. 磁场能量

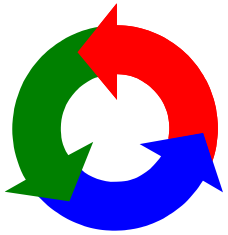
长直螺线管 $B = \mu \frac{N}{l} I$ $L = \frac{\psi}{I} = \frac{\mu N^2 S}{l}$

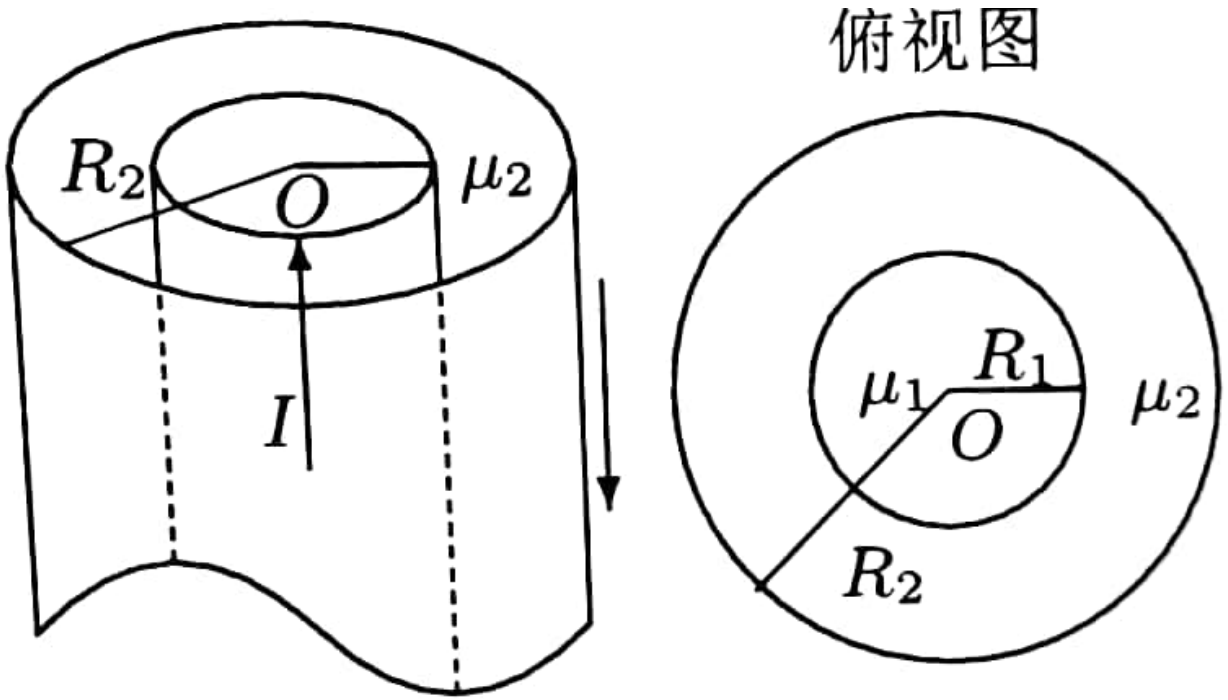
$$W_L = \frac{1}{2} L I^2 = \frac{1}{2\mu} B^2 V$$

$$\boxed{w_m = \frac{1}{2\mu} \vec{B}^2 = \frac{1}{2} \mu \vec{H}^2 = \frac{1}{2} \vec{B} \cdot \vec{H}}$$

在电磁场中 普遍适用

$$\boxed{w = \frac{1}{2} \vec{D} \cdot \vec{E} + \frac{1}{2} \vec{B} \cdot \vec{H}}$$





单位长度的自感？用场能计算